



STAPPONE SUITCASE 12

USER MANUAL

STAPPONE Medical

STAPPONE Physio

STAPPONE Research

1	Device description	5
2	stapp one Quick Start	7
2.1	Find out Sole ID	7
2.2	Insert Insoles into your shoe	7
2.3	Connect Insoles with your electronic device.....	7
2.4	Collect data on recording screen	7
3	Device Specifications	8
3.1	Bluetooth specifications	8
3.2	Resolution	8
3.3	Pressure sensor position.....	8
3.4	IMU Sensor Orientation	9
3.5	Sole dimensions.....	9
4	Symbols and Labeling	10
5	Intended Use	12
6	Contraindications.....	13
7	Important safety and device information.....	14
8	Handling instructions of the Insoles	16
9	Wireless Charging	18
	Indicator Light definition.....	18
	Charging process charger.....	19
	Charging process stapp one suitcase	20
10	Environment.....	22
11	Radiation and safety distance.....	23
12	Reprocessing	27
13	Medical solutions Software.....	28
	General information and symbols	28
	Symbols	29
	STEP 1 – Installation	31
	STEP 2 - Login	32
	STEP 3 - Settings and configuration.....	32
	User settings	33
	Camera use for video recording	33
	Connect your insoles	34
	Main directory of client data	34
	Test performing institute	34

Language	34
Data configuration.....	34
STEP 4 - New recording	35
Live view.....	35
Video in Recording Screen	36
'Discard data on insole' button	36
Start recording.....	36
During recording.....	36
End recording.....	36
Data synchronization.....	36
Add / search client.....	37
Rating / comment aids	37
STEP 5 - Analysis screen	37
Video in Analysis screen.....	37
Time/pressure graph.....	38
Center of Pressure (COP) visualization	38
Basic Data	39
Quality of Movement	39
Pressure Distribution.....	39
Stance and Swing Phases	40
Visualization settings.....	40
'Incl. General Information' checkbox.....	40
STEP 6 - Record list	40
Edit and delete recordings	40
Search filter	40
13.1.1 Compare recordings	41
STEP 7 – Data export	41
13.1.2 PDF report	41
13.1.3 CSV file export.....	44
14 Other equipment	46
15 Information about the manufacturer.....	47

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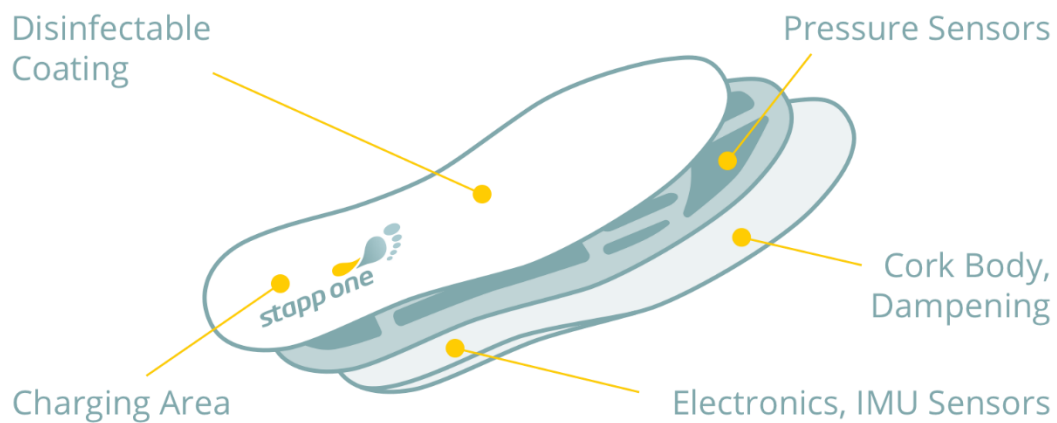
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1 DEVICE DESCRIPTION

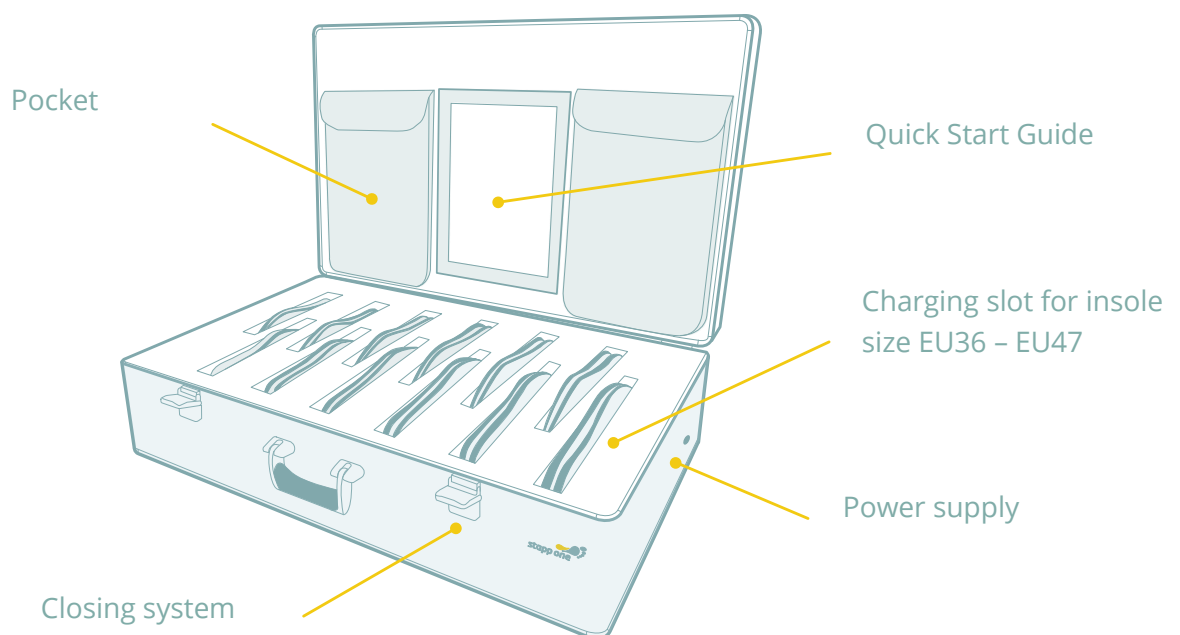
TWO SENSOR SOLES

This device is a sensor-based shoe insole, which measures movements relatively with a combination of pressure sensors and IMU sensors.



ONE CHARGING SUITCASE

The stappone sensorsoles can be charged via inductive charger integrated into the STAPPONE Suitcase 12. The STAPPONE Suitcase 12 contains 12 charging slots with two integrated inductive chargers each. In order to charge them the insoles need to be placed in the according slot.



In this manual the terms "stappone sensor insoles" and "device" refer to a pair of one left and one right sensor insole.

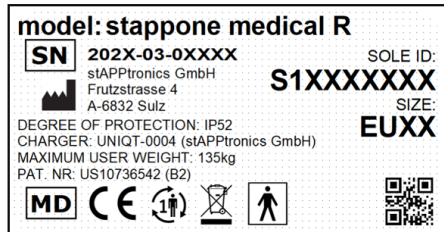
- The stappone sensor insoles are a medical device class I.
- For using this device no special qualification, except basic IT skills, is needed.
- This device is designed for 500 uses, or 1 million steps per device. One use is defined as one therapy unit.
- This device is designed for indoor- and outdoor-use in combination with closed shoes.
- This device is designed for Home Healthcare Environment & Professional Healthcare.
- The stappone sensor insoles are the applied part of the medical product

	Expected service life
Applied parts	
stappone sensor insoles	1 year
Accessories	
stappone wireless charger	1 year
USB-C Cable	1 year

2 stappone QUICK START

This process is a small overview of the key functionalities of the stappone system. Further information is available in the next chapters.

2.1 Find out Sole ID



On the product label of your insoles, you find the Sole ID. You need the Sole ID of the **right** insole to connect the insoles with the electronic device.

2.2 Insert Insoles into your shoe



If possible, remove the soles of your shoes you want to use the stappone sensor insoles with. Afterwards put the stappone sensor soles into your shoe.

2.3 Connect Insoles with your electronic device



To use the sensor insoles, make sure your devices integrated Bluetooth is activated. When logged in, connect your insoles (notice Sole ID) to your electronic device to by clicking on the 'CONNECT' button in setting screen.

2.4 Collect data on recording screen



To finally start data recording, go to recording screen. This data can be analysed immediately afterwards.

3 DEVICE SPECIFICATIONS

Sensors	12 pressure sensors per sole 2 acceleration sensors per sole 1 gyroscope per sole 1 magnet sensor per sole
Battery	Li-Ion battery cell with a capacity of 0.36 Ah per sole Approx. 10 hours running time
Charging	Wireless charging
Sizes	EU unisex sole sizes from 36 to 47

3.1 Bluetooth specifications

Frequency band	2.4GHz ISM Band (2.402 – 2.480 GHz utilized)
Channels	40 channels with 2 MHz spacing (3 advertising channels/37 data channels)
Max Tx Power	Class 2: 2.5 mW (+4 dBm)
Range	approx. 10 meters
Standard	Bluetooth Low Energy v4.2

3.2 Resolution

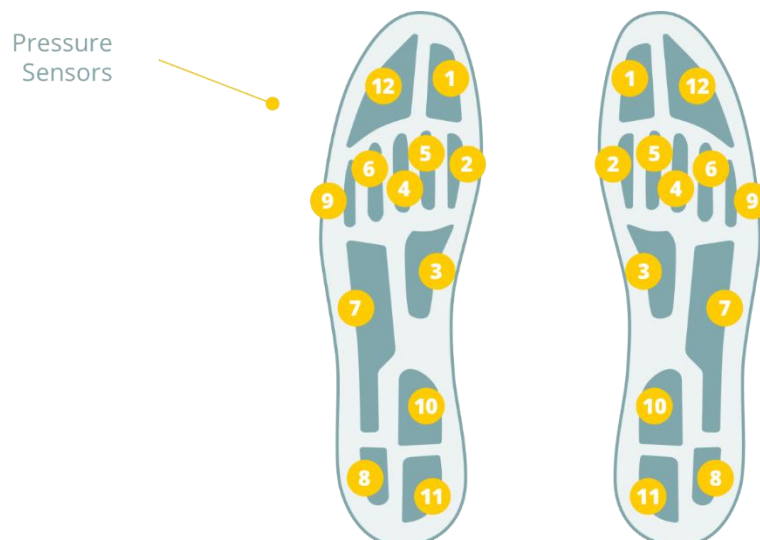
Pressure values: 0 - 1023 (relative pressure) – 10 Bit.

Accelometer values: -4000 to +4000 mg – 10Bit

Time resolution: 4 milliseconds

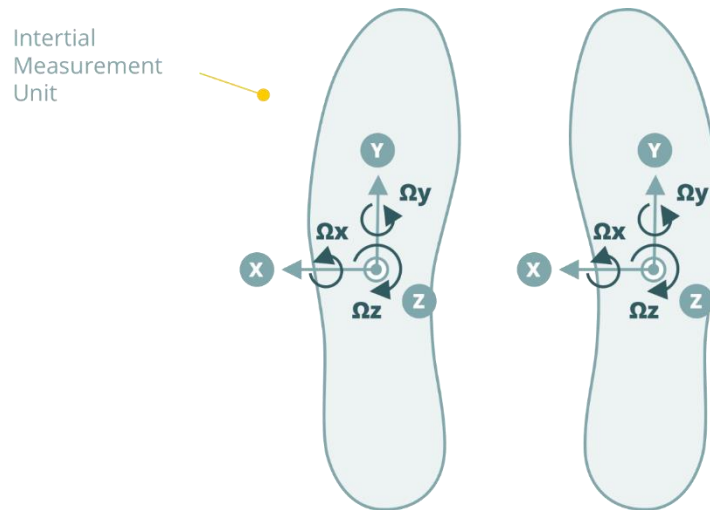
3.3 Pressure sensor position

Every insole has 12 integrated pressure sensors. The graphic below shows the position of every sensor. In the graphic the visualizations are equal to the left and right insoles.



3.4 IMU Sensor Orientation

X, Y and Z defines the orientation of the accelerometer and the magnet sensor.
 Ω_x , Ω_y and Ω_z defines the orientation of the gyroscope.
 In the graphic the left visualization is equal to left insole.



3.5 Sole dimensions

Sole size EUR	Length [mm]	Width [mm]
36	228,9	96,2
37	236,9	96,2
38	248,6	96,2
39	248,7	98,7
40	256,7	98,7
41	270,0	98,7
42	271,3	106,3
43	279,4	106,3
44	288,6	106,3
45	290,1	109,7
46	298,2	109,7
47	307,2	109,7

4 SYMBOLS AND LABELING

The label is located on the bottom side of both insoles as well as on the packaging.

Symbol	Definition
	This device is a medical device class I
	CE certificated
	Device serial Number
	Manual must be read before using this device
	The symbol on the device or its packaging signifies that this device must be disposed separately from ordinary household wastes at its end of life because it contains a battery and is electronic equipment. Ensure that you dispose of your stappone device at a recycling center.
	To identify a type BF applied part complying with IEC 60601-1
	CLASS II equipment complying with IEC 60601-1
	Direct current
	Prohibited: situations that could cause injury to yourself or others or damage your devices or other equipment.

	Warning: situations that could cause injury to yourself or others or damage your devices or other equipment.
	Mandatory: situations that could cause injury to yourself or others or damage your devices or other equipment.
	Name and address of the manufacturer (section 15)

5 INTENDED USE

- ❗ The following section is a simplified summary of the medical purpose statement.

This device is a sensor-based shoe insole, which measures the movements of our clients (patients or healthy persons) relatively with a combination of pressure sensors, acceleration sensors, gyroscopes and magnet sensors for diagnosis, therapy, and prevention. These relative data can be used for warning our clients if a pre-adjusted pressure limit is exceeded. The intended operators are clients and people with special education in health studies. Both intended operator groups can safely use all functions explained in this manual.

6 CONTRAINDICATIONS

The following conditions are absolute contraindications for using the device:

- ⊘ open wounds on the foot, irritated or otherwise unhealthy foot skin
- ⊘ severe gait impairments where the device might increase the risk of fall
- ⊘ inability of the client to wear closed shoes in combination with the stappone sensor insoles
- ⊘ body weight over 135 kg

When applying the sensor insoles to a client for more than one day, the user applying the device to the client must ensure that the wearer will not encounter negative orthopaedic long-term effects from wearing the device, and must instruct the client to report discomfort and pain due to wearing the device, and to stop wearing the sensor insole in such case.

7 IMPORTANT SAFETY AND DEVICE INFORMATION

Prohibited Action:

- ⊘ Do not wear your stapp one device while charging it.
- ⊘ Do not charge your stapp one device while it is wet.
- ⊘ Do not dispose of your stapp one device in a fire. The battery could explode.
- ⊘ Do not use your stapp one device in a sauna or steam room.
- ⊘ Do not use abrasive cleaners to clean your stapp one device.
- ⊘ Do not expose your stapp one device to extremely high or low temperatures.
- ⊘ Do not attempt to replace the battery or disassemble your stapp one device. Doing so will void the warranty and can result in a safety hazard.
- ⊘ Do not use your stapp one device if the stiffener is cracked.
- ⊘ Do not modify this equipment without authorization of the manufacturer.

Safety critical warning:

- ⚠ The device may cause skin irritation.
- ⚠ Prolonged contact may contribute to skin irritation or allergies in some users. To reduce irritation, follow three simple wear and care tips: (1) Keep it clean; (2) keep it dry; (3) let it breath.
- ⚠ Only use this device when your feet match the shoes in combination with the stapp one sensor insoles.
- ⚠ This device cannot be used by persons (including children aged 8 and above) with reduced physical, sensory or mental capabilities or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety, and understand the resulting dangers.
- ⚠ Consult your doctor before beginning or modifying any exercise program.
- ⚠ Consult your doctor before use if you have any pre-existing conditions that might be affected by your use of this stapp one device.
- ⚠ This device is not a toy. Do not allow children or pets to play with your stapp one device. The device contains small components that can be a choking hazard.
- ⚠ Substances in this device and its battery may harm the environment or cause injury if handled and disposed of improperly.
- ⚠ This device is not designed for Emergency Medical Service or use in critical environments (Emergency rooms, ICU etc.).
- ⚠ When using this product concern the information in section 10 regarding stacking and location to other equipment.

Mandatory action:

- ❗ Only use this device with socks
- ❗ The device contains electrical equipment that could cause injury if not handled properly
- ❗ Remove your stapp one device if it feels unusually warm or hot.
- ❗ Charge the battery in accordance with the instructions provided during setup. Only use a computer, powered hub or power supply that is verified by a recognized testing laboratory and an authorized stapp one charging device.

We urge everyone to see a doctor immediately if your health issues include back pain and additionally two of the following cases:

Bad general condition, known tumour disease, adequate trauma, osteoporosis, pronounced neurological deficits (loss of power or skin sensibility...), fever, night pain, uncontrolled departure of urine or stool, unintentional weight loss

8 HANDLING INSTRUCTIONS OF THE INSOLES



If possible, remove the soles of your shoes you want to use the stapp one sensor insoles with.



Find out the sole ID of the **right** insole you want to use (device label on the bottom side of the soles).



Put the stapp one sensor insoles into your shoe.



Make sure you put the stapp one sensor insoles in your shoes correctly (you are still able to see the stapp one logo).



Check if the stapp one sensor insoles are in full contact with the shoes.



Check if the stapp one sensor insoles matches the size of your shoes (if not try another size).



Put on your shoes.



Make sure the shoes with the stapp one sensor insoles match your feet and you have a stable stance in it.



Take off your shoes.



To remove the stapp one sensor insoles, lift sensor insole in medial mid foot area.



Pull on the heel area of the stapp one sensor insoles to remove it from your shoe.

9 WIRELESS CHARGING

Warnings:

- ⊘ Do not place foreign materials, such as metal objects, magnets, and magnetic stripe cards, between the insole and the wireless charger or the stappone suitcase. The insole may not charge properly or may be damaged due to overheating.
- ⊘ Do not use the wireless charger or stapp one suitcase with other devices than your stapp one sensor insoles. Do not try to recharge non-rechargeable batteries.
- ⊘ Do not touch insoles longer than 1 minute while charging.
- ⊘ Do not touch charger longer than 10 seconds while charging.
- ⚠ The charger or the stapp one suitcase **MUST** be placed outside the patient's environment (minimum distance of 1.5 meters), even when the device is not charging.
- ⚠ Wait 10 minutes after charging, to cool down your insoles, before using them.
- ⚠ While charging you may lose the Bluetooth connection to your insoles.

Indicator Light definition



A flashing blue light means that the charger is not supplied with sufficient power and the stapp one sensor soles cannot be charged without problems.



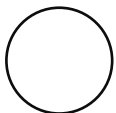
A constant blue light indicates that charging is in progress.



A constantly illuminated red indicator light signals an error in the process. Please remove the wireless charger from the insole and ensure that there are no obstacles between the wireless charger and the insole. Then align the wireless charger again as before and continue the charging process



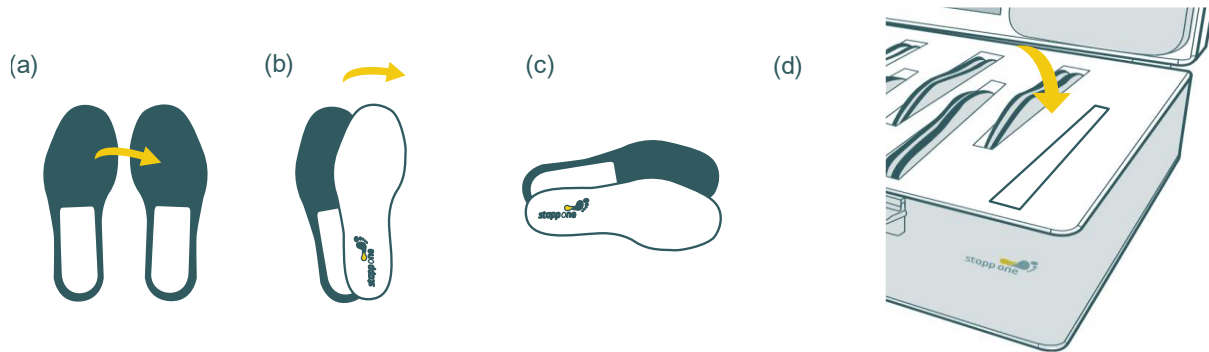
A flashing red indicator light signals that the charger has finished charging due to the high temperature and is cooling down. It is not necessary to remove the charger. This is a safety mechanism and not a cause for concern.



If the power lamp does not light up, it means that the charger is in standby mode and the sensor sole is not being charged. If the power lamp does not light up even though you have inserted the soles into the charging slot, please align the soles again. The heel area should be pressed completely to the edge so that the sensor soles are optimally placed in the charging slot.

Charging process charger

The stappone sensor soles are charged by induction - accordingly, to charge the soles, simply insert them into the corresponding charging slot (EU36 - EU47). The sensor soles must be inserted into the charging slot corresponding to the size according to the illustration below.



- (a) Place the sensor soles against each other with the bottom side facing inwards.
 - (b) Make sure that the stappone logo is facing outward.
 - (c) Make sure that the lateral side of the sensor soles is facing up.
 - (d) Insert the sensor soles into the appropriate loading slot. The sensor soles must be inserted with the heel area in the charger area.
- ❗ Be sure to press the sensor soles fully and firmly into the charging slot. Orient the soles in the charging slot until the blue status lamp lights up.

Warnings:

- ⚠ Only connect the USB charger to a properly installed grounded socket-outlet. The socket-outlet shall be installed near to the equipment and shall be accessible easily.
- ⚠ Use only USB chargers with an output voltage of 5V and a minimum of 1 A output current (SELV – Safe Extra Low Voltage).
- ⚠ Use only IEC 60601-1 certificated chargers.

1. For a full usage, both batteries (left and right insole) must be charged
2. To charge your stappone sensor insoles, you do not necessarily have to remove them from your shoes
3. Connect a USB C cable to the wireless charger
4. Place the wireless charger on a stappone sensor insoles so that the center of the charger is aligned with the heel of the insole
5. Move the charger slowly from the heel towards the toe area until the blue indication light starts to blink

Additional charging information:

If the indicator light does not work as described, disconnect the USB cable from the wireless charger and reconnect it.

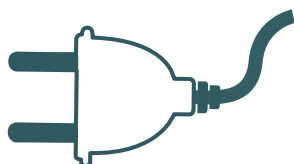
To save energy, unplug the USB charger when not in usage. The wireless charger does not have a power switch, so you must unplug the USB charger from the electric socket when not in use to avoid wasting power. The USB charger should remain closely to the electric socket and easily accessible while charging. If you have any technical problems, do not hesitate to contact us on our email support@stappone.com

Charging process stapp one suitcase

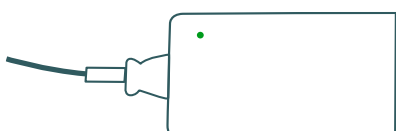
Warnings:

- ⊘ Do not use other power supplies than 'Mean Well Medical Adaptor GSM160A12' (Input: 100-240VAC, 50/60hz, 1.0-2.0A; Output: 12VDC, 11.5A)
- ⊘ Do not modify this equipment without authorization of the manufacturer
- ⚠ Apply disinfectant on the top area of the inlay regularly
- ⚠ Use the storage compartment on the inside of the suitcase whenever the Mean Well Medical Adaptor is not connected to the suitcase
- ❗ Only charge pairs of insoles with the stapp one suitcase

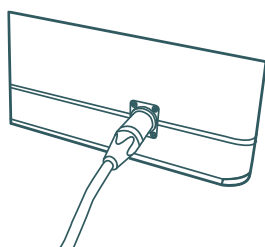
The stappone sensor soles are charged by induction - accordingly, to charge the soles, simply insert them into the corresponding charging slot (EU36 - EU47). The sensor soles must be inserted into the charging slot corresponding to the size according to the illustration below.



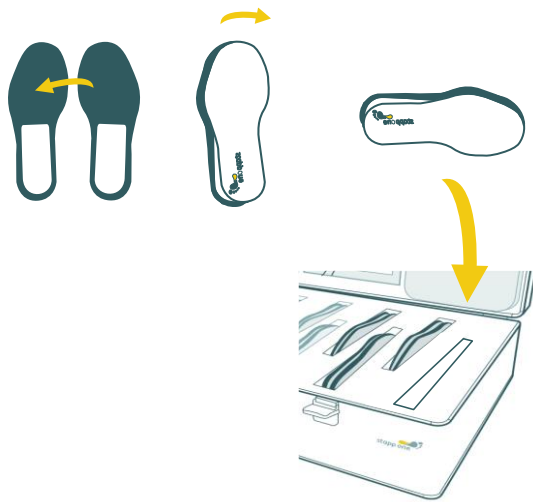
1. Connect power supply cable to an electric socket (100-240VAC, 50/60hz, 1.0-2.0A)



2. Connect power supply cable to Mean Well Medical Adaptor (constant green light signals, the cable is connected correctly)



3. Connect Mean Well Medical Adaptor to stapp one suitcase 12



4. Place the corresponding sensor sole pair in the loading slot provided. Make sure that all sensor sole pairs are inserted into the case according to the illustration.

- ❗ Proper storage in the case is important to ensure an unhindered charging process. If the chargers do not glow blue constantly, they must be realigned in the charging slot - press the soles deeply into the charging slot.

Additional charging information:

If the indicator light does not work as described, disconnect power supply from the electric socket and reconnect it.

If you have any technical problems, do not hesitate to contact us on our email support@stappone.com

10 ENVIRONMENT

Warnings:

- ⊘ The sensor insole must not be used in environments where radio emissions at 2.4 GHz are potentially harmful or forbidden by law.
- ⚠ Disadvantageous radio performance may occur if used in an environment with other devices operating at 2.4 GHz (e.g. WiFi, Bluetooth). In this case, stop using the sensor insole in this environment.

Environment Specifications	
Operation humidity range	15 – 90 [%rH], noncondensing
Storage and transportation humidity range	5 – 90 [%rH], noncondensing
Air pressure range for operation	< 3000m (> 690hPa)
Air pressure range for storage	< 13000m (> 190hPa)
Operation temperature	5 – 40 C°
Storage and transportation temperature	-25 – 70 C°
Time required for ME (medical electrical) equipment to warm from the minimum storage temperature between uses until it is ready for intended use	1 hour (room temperature)
Time required for me equipment to cool from the maximum storage temperature between uses until it is ready for intended use	1 hour (room temperature)
Degree of protection (IEC 60529)	IP52 (Full protection against contact, protection against interior injurious dust deposit and protection against diagonal water drips)

11 RADIATION AND SAFETY DISTANCE

Warnings:

- ⚠ Other cables and accessories may negatively affect EMC performance.
- ⚠ Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30 cm (12 inches) to any part of the stapp one medical, including cables specified by the manufacturer.

For safety reasons, the following tables must be observed with regard to EMC (Electromagnetic Compatibility). They are part of the certification according to the standard EN 60601-1-2 and taken from it accordingly.

Portable and mobile RF communications equipment (radios) should not be used closer to any part of the 'stapp one medical' product than the recommended safety distance specified in the tables below. In the following tables ME device is used for 'Medical electrical device' and describes the device 'stapp one medical'

Accessories	Type	Cable length
2x USB Cable	C	150 cm
2x stapp one wireless charger	UNIQT-0004	N/A
stapp one suitcase	N/A	N/A

Electromagnetic emissions: manufacturer's declaration and guidelines

(Table according to EN60601-1-2)

Guidance and manufacturer's declaration - electromagnetic emissions		
The ME device is intended for use in an electromagnetic environment as specified below. The user of the ME device should ensure that he is operating in such an environment.		
EMI measurements	Conformity	Electromagnetic environment - guidelines
RF emissions to CISPR 11	Group 1	The ME device uses RF energy only for its internal function. Therefore, its RF transmission is very low and it is unlikely that neighbouring electronic devices will be disturbed. The ME device is suitable for use in facilities other than residential and those directly connected to the public utility network that also supplies buildings used for residential purposes.
RF emissions to CISPR 11	Class B	
Transmission of harmonics according to IEC 61000-3-2	N/A	
Transmission of voltage fluctuations / flicker according to IEC 61000-3-2	N/A	

Electromagnetic immunity: manufacturer's declaration and guidelines

(Table according to EN 60601-1-2)

Guidance and manufacturer's declaration - electromagnetic immunity			
The ME device is intended for use in an electromagnetic environment as specified below. The user of the ME device should ensure that he is operating in such an environment.			
Immunity	IEC 60601-1-2:4 th Edition	Matching level	Electromagnetic environment - guidelines
Static electricity discharge (ESD) according to IEC 6100-4-2	± 8 kV contact discharge ± 15 kV air discharge	± 8 kV contact discharge ± 15 kV air discharge	Floors should be wood, concrete or ceramic tile. If the floor is covered with synthetic material, the relative humidity must be at least 30%.
EFT/Burst according to IEC 61000-4-4	± 2 kV - AC Mains 100 kHz PRF	N/A	The quality of the supply voltage should be that of atypical business or hospital environment.
Surges according to IEC 61000-4-5	± 1 kV Voltage outer conductor to outer conductor	N/A	The quality of the supply voltage should be that of a typical business or hospital environment.
Voltage dips, short-term interruptions and fluctuations in the supply voltage according to IEC 6100-4-11	UT = 0%, 0.5 cycle (0, 45, 90, 135, 180, 225, 270 and 315°) UT = 0 %; 1 cycle UT = 70%; 25/30 cycles (@ 0 degrees) UT = 0%; 250/300 cycle	N/A	The quality of the supply voltage should be that of atypical business or hospital environment. If the user of the ME device requires continued operation even in the event of power interruptions, it is recommended that the foot scanner be powered from an uninterruptible power supply or a battery.
Magnetic field at the supply frequency (50 / 60Hz) according IEC 61000-4-8	30 A/m	30 A/m	Mains frequency magnetic fields should be the typical value found in the business and hospital environment.
Note: U_T is the mains AC voltage before the application of the test levels.			

Electromagnetic Immunity: Manufacturer's Declaration and Guidelines for Non-Life-Causing ME Equipment

(Table according to EN 60601-1-2)

Guidance and manufacturer's declaration - electromagnetic immunity			
The ME device is intended for use in an electromagnetic environment as specified below. The user of the ME device should ensure that he is operating in such an environment.			
Immunity	IEC 60601 test level	Matching level	Electromagnetic environment - guidelines
Conducted RF interference according to IEC 61000-4-6	3V 150kHz to 80MHz ISM Bands between 0,15 MHz and 80 MHz 80% AM at 1 kHz	N/A	Portable and mobile radios should be used no closer to the ME device, including the wires, than the recommended safe distance calculated using the equation applicable to the transmit frequency. Recommended safety distance in meters: $d = (3,5/3) \cdot \sqrt{P}$ $d = (3,5/3) \cdot \sqrt{P}$ 80MHz to 800MHz $d = (7/3) \cdot \sqrt{P}$ 800MHz to 2,5GHz with P as the nominal power of the transmitter in watts (W) according to the transmitter manufacturer and d as the recommended safety distance in meters (m). The field strength of stationary radio transmitters should be lower than the compliance level at all frequencies according to an onsite survey. In the vicinity of devices bearing the following icon, interference is possible.
Radiated RF interference according to IEC 61000-4-3	10V/m 80MHz to 2,7GHz 80% AM at 1 kHz	10V/m 80MHz to 2,7GHz 80% AM at 1 kHz	
Proximity Field from Wireless Transmitters	9 V/m to 28 V/m 15 specific frequencies	28 V/m	Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30 cm (12 inches) to any part of the stapp one medical including cables specified by the manufacturer.
Electrical Transients – Vehicle 12 Volt Powered	ISO 7637-2 Pulses – 600V max.	N/A	
Note 1: At 80 MHz and 800 MHz, the higher frequency range applies. Note 2: These guidelines may not be applicable in all cases. The spread of electromagnetic quantities is influenced by absorption and reflection of buildings, objects and people.			

- a The field strength of stationary transmitters, e.g. Base stations of radio telephones and land mobile radios, amateur radio stations, AM and FM radio and television stations can not be theoretically accurately predicted. In order to determine the electromagnetic environment with respect to the stationary transmitters, a study of the electromagnetic phenomena of the location should be considered. If the measured field strength at the location where the ME device is used exceeds the above compliance levels, the ME device should be observed to verify its intended function. If unusual performance is observed, additional measures may be required, such as: a changed orientation or another location of the ME device.
- b Over the frequency range of 150kHz to 80MHz the field strength should be less than 3V / m.

Recommended separation distances between portable and mobile HF telecommunications equipment and the ME device

(Table according to EN 60601-1-2)

Recommended safety distances between portable and mobile HF telecommunications equipment and the ME device			
The ME device is intended for operation in an electromagnetic environment in which the RF disturbances are controlled. The user of the ME device may thereby help to avoid electromagnetic interference by maintaining the minimum distance between portable and mobile RF telecommunications equipment (transmitters) and the ME equipment, depending on the output power of the communication equipment as indicated below.			
Rated power P of the transmitter W	Guard distance, depending on the transmission frequency m		
	150kHz to 80MHz $d=(3,5/3)*\sqrt{P}$	80 MHz to 800MHz $d=(3,5/3)*\sqrt{P}$	800MHz to 2,5GHz $d=(7/3)*\sqrt{P}$
0,01	0,12	0,12	0,23
0,1	0,37	0,37	0,74
1	1,17	1,17	2,33
10	3,69	3,69	7,38
100	11,67	11,67	23,33
For transmitters whose maximum rated power is not specified in the table above, the recommended guard distance d in meters (m) can be obtained using the equation associated with each column, where P is the transmitter maximum power rating in watts (W) as specified of the transmitter manufacturer.			
Note 1: At 80 MHz and 800 MHz, the higher frequency range applies.			
Note 2: These guidelines may not be applicable in all cases. The spread of electromagnetic quantities is influenced by absorption and reflection of buildings, objects and people.			

12 REPROCESSING

Warnings:

- ⚠ The stappone sensor insoles are no sterile device
- ❗ Use disinfectant for cleaning the device
- ❗ To ensure the function of your insoles they must be charged regularly
- ❗ If worn by different wearers, the sensor insole must be reprocessed before it can be reused

Process:

This process must be done for the first and each subsequent use of insoles:

1. Take the insoles out of your shoes
2. Apply disinfectant on the top active area
3. Dry the top active area with a wipe
4. Check if the device has obvious damage
5. Put the insoles back into your shoes
6. If you have any problems, do not hesitate to contact us on our email support@stappone.com

13 MEDICAL SOLUTIONS SOFTWARE

Warning:

- ⚠ The software is only visualizing sensor data. No diagnosis is given.
- ⚠ For the use of the software a Windows capable end device with software version Windows 10 (or higher) is necessary

General information and symbols

Our product includes three different software solutions that are available for customers to purchase. The software solutions are all used for data visualization and recording in the context of a clinical setting with, on the one hand, the treating medical staff and, on the other hand, patients with a corresponding (medical) indication as direct users. The software variants differ only in their context of use and the available features. These features are described below. In the following, the features and functions are described in detail, mentioning accordingly which software solution has the corresponding function.

	Physio	Medical	Research
			
Quick installation and easy to use	✓	✓	✓
Patient database	✓	✓	✓
Life pressure distribution visualization	✓	✓	✓
Live COP (center of pressure)	✓	✓	✓
Video analysis	✓	✓	✓
Motion analysis feature	✓	✓	✓
Assessment tests	✓	✓	✓
PDF analysis report	✓	✓	✓
Raw data export			✓
Sampling rate adaptable			✓

Symbols

Symbol	Where	Definition
	Header	Button to view manual
	Header	Loading status insoles. A textbox appears when moving cursor on this symbol
	Header	No insoles connected
	Header	Button to navigate to setting screen
	Header	Button to logout current user
	Header	Recording running
	Header	Data is synchronizing
	Header	Data synchronized successfully
	Setting Screen	Soles connected
	Setting Screen	User settings

	Setting Screen	Button to refresh
	Recording Screen	Button to start recording
	Recording Screen	Button to stop recording
	Analysis Screen	Button to link two recordings while comparing
	Recording Screen	Button to delete recording
	Recording Screen	Button to edit recording

STEP 1 – Installation

Research



Available

Medical



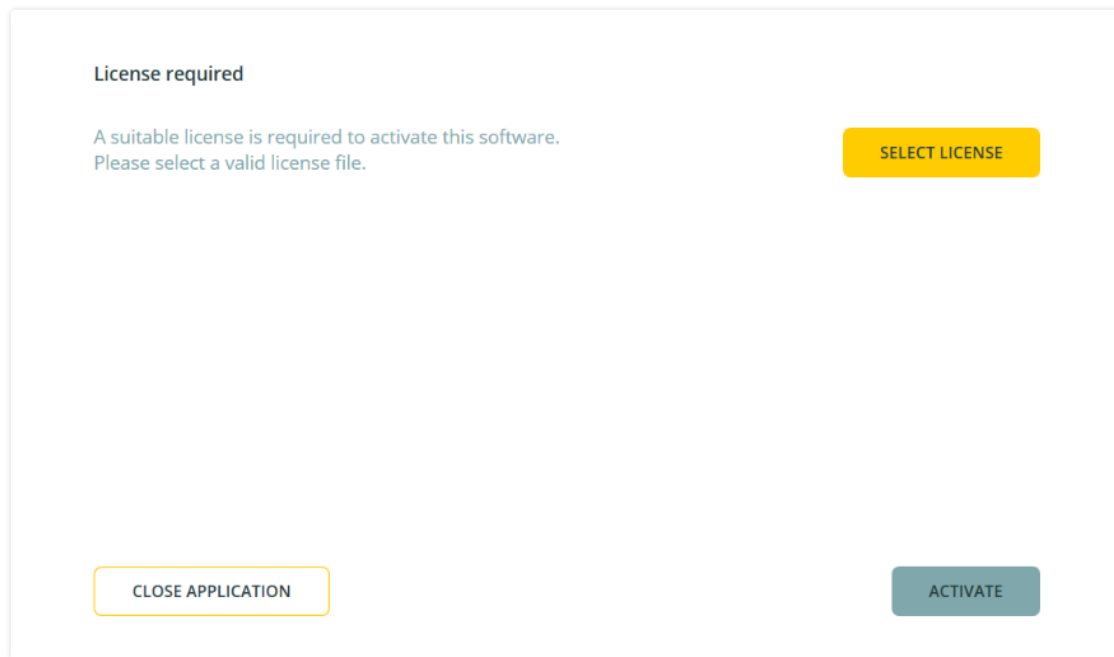
Available

Physio

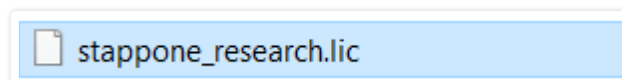


Available

When the software is started for the first time, the following popup appears. Click on the 'SELECT LICENSE' button.



Then select the license file which is delivered as part of the product purchase. Afterwards click the 'ACTIVATE' button.



STEP 2 - Login

Research



Available

Medical



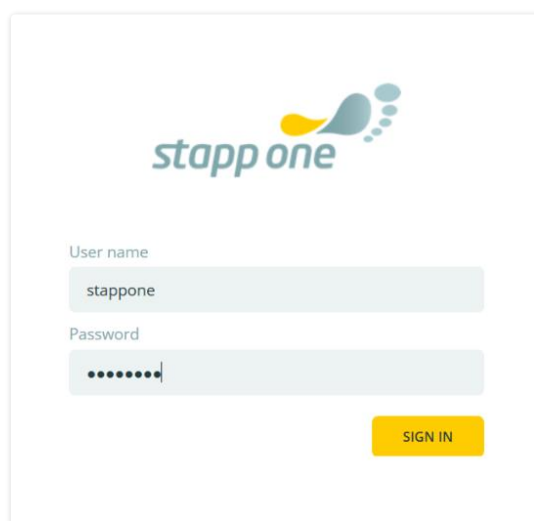
Available

Physio



Available

For the first login use default username 'stappone' and password 'stappone'.



The login form displays the 'stapp one' logo at the top. Below it, there are two input fields: 'User name' with the text 'stappone' and 'Password' with masked characters '.....'. A yellow 'SIGN IN' button is located at the bottom right of the form.

STEP 3 - Settings and configuration

Research



Available

Medical



Partly available

Physio



Partly available

Warnings:

- ❗ Both Insoles must be charged and activated for a connection to be able to be established
- ❗ When connecting the insoles to the device, make sure to minimize the distance between the insoles and the device

User settings

It is possible to create users and administrators. Therefore, switch to user settings by clicking on this symbol in the setting screen:



To delete the default user with the default password, create a new administrator. Then log out by clicking on this button:



Then login with your new administrator account. Afterwards delete the default stappone user.

Make sure not to forget the passwords of all administrator users. Passwords cannot be reset without an administrator user being able to login. Resetting all users with a fresh installation can lead to data loss.

	Administrator	User
Login	✓	✓
Change own password	✓	✓
All analysis and recording features available in the software	✓	✓
Change password of another Administrator/ User	✓	
Create new Administrator/ User	✓	
Delete Administrator/ User	✓	
Enable/Disable Administrator/ User	✓	

Camera use for video recording

Select the camera you want to use. It is possible to use common webcams connected via USB or any camera which provides COM port access. If there is no need for a camera, select 'none'.

Camera use for video recording

None

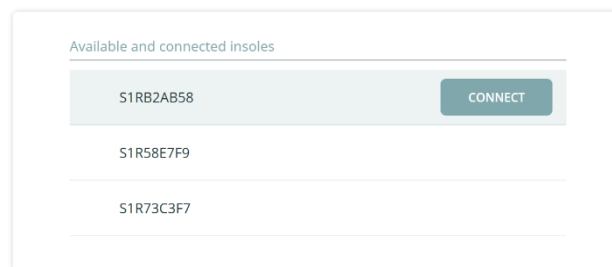
↺

Connect your insoles

Warnings:

- ❗ It is only possible to connect one pair of insoles at the same time
- ❗ Insoles should be within 5 meters of the device to ensure reasonable connectivity
- ❗ Shake both insoles or give them a slight tap before connecting to ensure they are active
- ❗ Make sure your devices Bluetooth is turned on

Before connecting the insoles, make sure your devices integrated Bluetooth is on. Select 'CONNECT' button of the sole ID of the **right** insole you want use (found on the insole label) in section 'Available and connected insoles'. If there are no insoles shown, try to turn your Bluetooth off and on again.



Main directory of client data

Choose your directory with the button 'EDIT' in section 'Main directory of client data'.

Test performing institute

Add information about your company/institute. This information is going to be added to your reports.

Language

Choose between English and German.

Data configuration



Warning:

- ⚠ A higher sampling frequency leads to longer transmission durations

In this section it is possible to set the sampling frequency. You can choose which data you want to record (pressure and/or accelerometer). Only available in Research Version.

STEP 4 - New recording

Research



Available

Medical



Available

Physio



Available

After all settings are configured and you are still on the setting screen, switch to the Recording screen via pressing 'RECORD LIST' button in the left top corner. Then click on the 'NEW RECORDING' button. INFO: In the picture below no video camera has been chosen.



Live view

Warnings:

- ⚠ Due to dynamic calibration in the live view the visualization of live pressure values could possibly be incorrect
- ❗ Walk a few steps with the insoles for more accurate live visualizations
- ❗ For correct analysis use record mode

In the Live View section in the Recording screen it is possible to see the live load distribution of left and right sole. Moreover, a COP for each insole is visualized (black). The red point is the COP of both insoles. The Live View Feature can be used for live feedback. A dynamic calibration is used for this feature.



Video in Recording Screen

This section is only available if a camera has been chosen in the setting screen. This feature can be used to align the camera before recording or to give the patient visual feedback.

'Discard data on insole' button

This button deletes all recorded data which is saved on the insoles. This might be useful if the synchronization process failed for the recorded data before.

Start recording

Warning:

- ⓘ At the time the recording should be started a Bluetooth connection between the electronic device and insoles is required

Click on the 'start recording' button to start recording.



During recording

During the recording no active Bluetooth connection is required. The data is saved on the memory which is inside the insoles.

End recording

Warning:

- ⓘ At the time the recording should be stopped a Bluetooth connection between the electronic device and insoles is required

To stop the recording use the 'stop recording' button.



Data synchronization

After ending the recording, the synchronisation starts automatically. The indicator for this process is a symbol in the upper right corner of the screen. The progress is measured in percent. When the sync process has ended a green saving-symbol appears.



Tipp: A shorter distance between insoles and device improves the synchronization duration.

Add / search client

To add a new client, click on the '+' Button in 'Client' section.

To find a client who is already registered in the system, enter the Name, ID or date of birth in the search field.

Client

+

Rating / comment aids

It is possible to add a rating, a comment and information about used aids. This information is stored with the recording.

STEP 5 - Analysis screen

Research



Available

Medical



Available

Physio



Available

In this screen you can analyse data of selected recording in the record screen. This screen opens automatically after you save a new recording. The arrow buttons in the top mid area of the screen can be used to navigate through recordings.

Video in Analysis screen

If the recorded video and the sensor data which is visualized in the time/pressure graph is not synchronic, an offset can be set via '-' and '+' buttons.

-

+

Video/Data Offset: 0.05 sec.

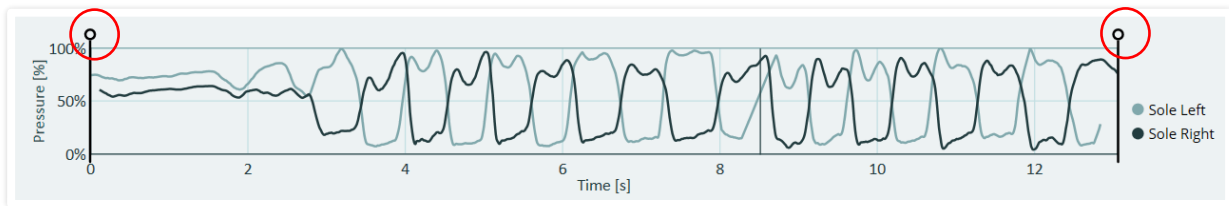
The dropdown function below can be used to manipulate speed of the video.

x1.0

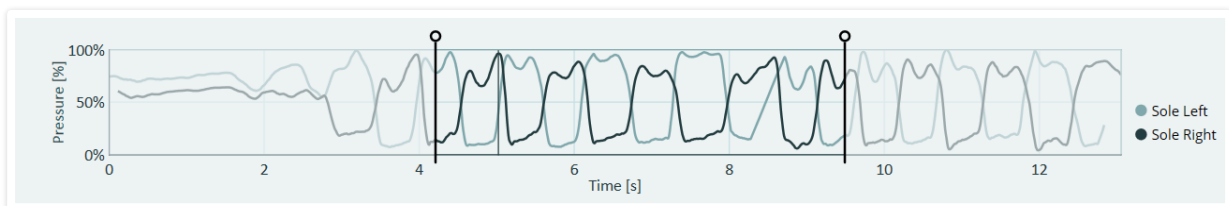
Time/pressure graph

Warning:

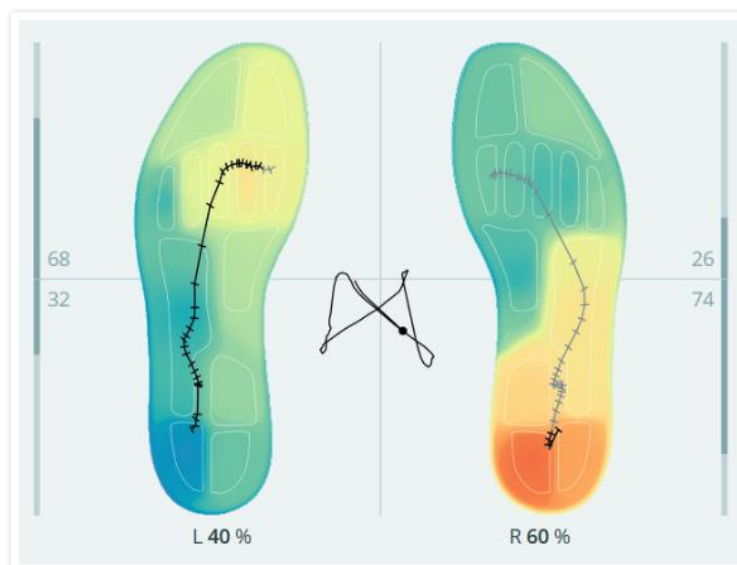
⚠ Calculated parameters change when changing the analysed area.



Aggregated relative pressure data of the sensor soles in percent for the left and right foot along the test duration are displayed in seconds. Via the red circled handlers, it is possible to limit the analysed area.



Center of Pressure (COP) visualization



Following parameters can be seen in the visualisation above

- Forefoot and hindfoot pressure distribution
- Left/right pressure distribution
- Centre of Pressure visualization
- Cyclogram (see section 13.1.2.1 Explanation of all relevant)

Basic Data

Duration

Time that was required for the entire test run.

Number of steps

Number of steps during the test run.

Distance

This field is to set the distance the patient walked during recording. If the distance is edited afterwards the appearing 'save' button in the lower right edge must be clicked.

Double stride length

Average distance covered from one touchdown to the next of the same foot over the entire test duration in relation to the specified distance.

Speed

Average gait speed over the duration of the test run based on the specified distance.

Cadence

Frequency in steps per minute (spm).

Gait cycle

The interval of time between the point when the foot initially contacts the ground, to when the same foot contacts the ground again.

Double Support Time

Time in which both feet made ground contact at the same time in an averaged gait cycle phase.

Quality of Movement

Symmetry index

Describes how rhythmically a gait cycle takes place. For calculation, ground contact time and foot pressure of the left and right leg are compared. A physiological gait has the distribution 50% to 50 % with a possible deviation of 5%.

Regularity

The value of this parameter depends on the regularity of the stand phase times of a recording. To have a more reliable result, more than 20 steps are recommended.

Pressure Distribution

Overall

Percentage pressure load of both sensor soles to each other.

Inside / medial

Percentage pressure on the inside of the foot compared to the outside.

Outer / lateral

Percentage pressure load on the outside of the foot compared to the inside.

Forefoot

Percentage pressure load on the forefoot compared to the rearfoot.

Hindfoot

Percentage pressure load on the rearfoot compared to the forefoot.

Stance and Swing Phases

Stance phase

Averaged proportion of ground contact times in the gait cycle.

Swing phase

Average proportion of times without contact with the ground in the gait cycle.

Visualization settings

This section can be used to activate/deactivate visualization features.

'Incl. General Information' checkbox

If this checkbox is activated, an appendix is attached to the PDF report, where all relevant parameters are defined.

STEP 6 - Record list

Research



Available

Medical



Available

Physio



Available

All recorded data is listed in 'record list'. The newest recording is on top of the list.

Edit and delete recordings

To edit/delete recordings, the cursor must be over the relevant recording. Symbols for editing and deleting appear.

Search filter

To find recordings of a client who is already registered in the system, enter the name, ID or date of birth.

Suche / Filter

13.1.1 Compare recordings

It is possible to compare two recordings. In order to compare two recordings, they must be chosen by clicking on the checkboxes in the relevant rows. The recording with the left activated checkbox is going to be on the left side of the Analysis screen. The recording with the right activated checkbox is going to be on the right side of the Analysis screen.

No.	Compare	Recorded	Client, ID, Date of birth	Comment
003	☒ ☒	08.06.2020, 13:54	No client assigned to data set.	
002	☒ ☒	08.06.2020, 08:20	No client assigned to data set.	
001	☒ ☒	21.01.2020, 14:33	No client assigned to data set.	

In the Analysis screen this symbol can be used to link the two compared recordings to each other.



STEP 7 – Data export

Research



Available

Medical



Not available

Physio



Not available

13.1.2 PDF report

To create a report click on the 'PDF EXPORT' button.

13.1.2.1 Explanation of all relevant terminology

In this section, the terminology and all parameters, which are useful to interpret the PDF Report, are explained.

Supervisor

Person who carried out the test.

Duration

Time that was required for the entire test run.

Speed

Average gait speed over the duration of the test run based on the specified distance.

Pressure-time diagram

Aggregated relative pressure data of the sensor soles in percent for the left and right foot along the test duration in seconds.

Number of steps

Number of steps during the test run.

Cadence

Frequency in steps per minute (spm).

Double stride length

Average distance covered from one heelstrike to the next of the same foot over the entire test duration in relation to the specified distance.

Gait cycle

The interval of time between the point when the foot initially contacts the ground, to when the same foot contacts the ground again.

Double support time

Time in which both feet made ground contact at the same time in an averaged gait cycle phase.

Stance phase

Average proportion of ground contact times in the gait cycle.

Swing phase

Average proportion of times without contact with the ground in the gait cycle.

Average of maximum pressure

Average values of the maximum pressure per step over the entire course of the test.

Cyclogram

Shows the COP of both sensor soles over time. A physiological gait pattern is represented by a characteristic "butterfly shape"

Total pressure

Percentage pressure load of both sensor soles to each other.

Inner / medial

Percentage pressure on the inside of the foot compared to the outside.

Outer / lateral

Percentage pressure load on the outside of the foot compared to the inside.

Forefoot

Percentage pressure load on the forefoot compared to the rearfoot.

Hindfoot

Percentage pressure load on the hindfoot compared to the forefoot.

Symmetry index

Describes how rhythmically a gait cycle takes place. For calculation, ground contact time and foot pressure of the left and right leg juxtaposed. A physiological gait has the distribution 50% to 50% with a possible deviation of 5%.

Ground contact time

Duration of the average contact of the foot on the ground per step.

Abbreviations

%	percent
cm	distance in centimeters
m	distance in meters
min	duration minutes
s	duration in seconds
spm	steps per minute

Glossary:

Gait speed

Speed is a key indicator of a person's ability to walk. It is the central measured value for the quantitative gait analysis. Persons instinctively choose the speed that is appropriate for their locomotor skills. A strong slowdown in speed is one sure indication of a pathological event. There are many gait parameters that are depending on the gait speed.

Standard values for the free gait speed of different age groups, women and men have been published in the literature over the past few decades.

The norm values for the physiological free gait speed of adults are in the following area:

Free gait velocity:

[m / sec] 1.20 - 1.50

[km / h] 4.32 - 5.40

Cadence (gait frequency)

The gait frequency or cadence is usually recorded in number of steps per minute. If the gait speed is increased, stride length and gait frequency/cadence usually increase proportionally. When slowing down, stride length and gait frequency decrease accordingly. This applies when the cadence lies approximately between 80 and 120 steps per minute.

When moving with a higher speed, the step length is limited by the leg length. Walking slowly with less than 80 steps per minute is hardly possible with a fluid, symmetrically alternating gait.

The normal values for adults are 105 to 130 steps / min.

Gait cycle

Walking is a cyclical movement that is repeated over and over again defining a motion sequence. This movement section is specified as a gait cycle. The gait cycle characterized as duration from the initial contact with the ground until the initial contact with the same leg. Consequently, a gait cycle can be defined for the right leg and for the left leg. Certain events occur within the gait cycle and therefore it makes sense to further differentiate between the stance and swing phase. Standard values for the duration of a gait cycle are about 1.0 seconds.

Double Support Time

A more subtle and easy way to determine the stance phase is the division into two phases. A distinction can be made between the two-legged stance phase and the one-legged stance phase. The two-legged stance phase is characterized by both feet having ground contact at the same time in an averaged gait cycle.

The main characteristic of the double support time is that both feet are in contact with the ground and for the one-legged standing phase only one foot touches the ground. The two-legged phases are those with the highest static stability. If the walking speed is reduced, the duration of the two-legged standing phases is extended accordingly. A disturbed balance or lack of stability usually lead to a slower walking speed. The double standing phase during a gait cycle is approximately 20-24% at medium walking speed. When walking very slowly, about 46% of the gait cycle will be attributed to the double standing phases.

COP or cyclogram

The COP is shown in the middle of the footprint graphic. It shows the centre of pressure of both sensor soles over time. This represents the course of the printing focus of the entire body in the form of lines. Usually these lines show a characteristic "butterfly shape" and correspond to the symmetry of the aisle.

Symmetry index

The symmetry index describes how rhythmically a gait cycle takes place. It compares the foot pressure of the left foot with the right foot. A balanced gait is described with L 50% and R 50%, whereby a deviation of +/- 5% points is considered not pathological.

13.1.3 CSV file export



To export the raw data press the 'DATA EXPORT' button. Only available in Research Version.

13.1.3.1 Pressure:

Sensor placement is visualized in section 3.3.

sole_id	timestamp	sv1	sv2	sv3	sv4	sv5	sv6	sv7	sv8	sv9	sv10	sv11	sv12
1	1,5839E+12	414	453	402	417	462	423	512	506	418	449	475	369
2	1,5839E+12	383	378	335	331	313	290	424	407	337	392	379	279
1	1,5839E+12	414	453	402	416	462	423	512	506	418	449	474	369
2	1,5839E+12	383	377	335	331	313	290	425	407	338	392	378	280
1	1,5839E+12	411	452	401	415	463	423	511	505	418	447	473	367
2	1,5839E+12	382	376	335	331	312	290	427	406	339	391	377	280

sole_id:

1 = right

2 = left

timestamp:

Unix Timestamp in milliseconds from 01.01.1970 (UTC).

sv1-12:

Pressure data from sensor 1 to 12.

13.1.3.2 Accelerometer:

Sensor orientation is visualized in section 3.4.

sole_id	timestamp	acc_x	acc_y	acc_z
1	1,5839E+12	96	-136	976
2	1,5839E+12	88	-128	968
1	1,5839E+12	96	-160	976
2	1,5839E+12	80	-112	976
1	1,5839E+12	104	-152	984
2	1,5839E+12	88	-112	968

sole_id:

1 = right

2 = left

timestamp:

Unix Timestamp in milliseconds from 01.01.1970 (UTC).

acc_x – acc_y – acc_z:

Accelerometer values in x-, y- and z-axis.

14 OTHER EQUIPMENT

To improve the reliability of your studies and recordings, stapp one offers test shoes. This shoes also extend the life span of your insoles.

To have a visual feedback a webcam is necessary.

With questions and concerns do not hesitate to contact us.

15 INFORMATION ABOUT THE MANUFACTURER



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STAPPONE Suitcase 12 User Manual DE 1.2

Creator

- Complete and correct content
- Formally correct execution

Role	Date	First and last name
Quality Manager	2023-09-08	Anja Buchberger

Reviewer

Checked

- Complete and correct content
- Plausible content
- If required, detailed criteria are fulfilled (see annex, if applicable)

Role	Date	First and last name
Head of Software	2023-09-11	Nicole Howard

Releaser

Released

- The document objective is met
- Creation and testing are completed
- If required, detailed criteria are fulfilled (see annex, if applicable)

Role	Date	First and last name
Managing Director	2023-09-11	Peter Krimmer

Document change history

Version	Date	Author	Status	Change description
0.4.1	2020-03-08	DS	Archived	First version
1.0.0	2021-10-25	AB	Archived	Translation to German and minor wording changes
1.1.0	2022-12-29	AB	Archived	Layout changes
1.2.0	2023-09-11	AB	Released	Integrated changes regarding Bluetooth

Table 1: Documents change history